

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Dot .Net Core and Progressive Web Application Practical

Practical – VII

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Class: TYIT	Batch: 1
Date of Assignment: 04/08/26	Date/Time of Submission: 04/8/26

a. Develop a PWA that works in offline mode using Service Worker.

Code:

<pre>Manifest.json { "name": "College Information Portal", "short_name": "CIP", "start_url": "/", "display": "standalone", "icon": [{ "src": "images/icon1.png", "size": "128x128", "type": "image/png", "purpose": "any" }, { "src": "images/icon2.png", "size": "512x512", "type": "image/png", "purpose": "any" }] }</pre>	<pre>service-worker.js self.addEventListener("install", e => { e.waitUntil(caches.open("v1").then(c => c.add("/offline.html"));); }); self.addEventListener("fetch", e => { e.respondWith(fetch(e.request).catch(() => caches.match("/offline.html"))); });</pre>	<pre>Layout.cshtml <link rel="manifest" href="/manifest.json"/> <script> if('serviceWorker' in navigator){ navigator.serviceWorker.register('/service- worker.js'); } </script></pre>
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Output:

You are currently offline

Please check your internet connection and try again.

B. Create a Student Record Management PWA.

Code:

```

service-worker.js
const CACHE = "v1";

self.addEventListener("install", e => {
  e.waitUntil(
    caches.open(CACHE).then(cache => {
      return cache.addAll([
        "/",
        "/Home/Index",
        "/css/site.css",
        "/js/site.js"
      ]);
    })
  );
});

self.addEventListener("fetch", e => {
  e.respondWith(
    caches.match(e.request).then(response => {
      return response || fetch(e.request);
    })
  );
});

Manifest.json
{
  "name": "College Information Portal",
  "short_name": "CIP",
  "start_url": "/",
  "display": "standalone",
  "icon": [
    {
      "src": "images/icon1.png",
      "size": "128x128",
      "type": "image/png",
      "purpose": "any"
    },
    {
      "src": "images/icon2.png",
      "size": "512x512",
      "type": "image/png",
      "purpose": "any"
    }
  ]
}

```

```

Student.cs
namespace HTML_in_offline.Models
{
  public class Student
  {
    public int RollNo { get; set; }
    public string Name { get; set; }
  }
}

```

```

HomeController.cs
public IActionResult Index()
{
  Student s = new Student();

  s.RollNo = 101;
  s.Name = "Rahul";

  return View(s);
}

```

```

Index.cshtml
@{
  ViewData["Title"] = "Home Page";
}

<div class="text-center">
  <h1 class="display-4">Welcome</h1>
  <p>Learn about <a
href="https://learn.microsoft.com/aspnet/core">building Web
apps with ASP.NET Core</a>.</p>
  <h1></h1>
  <h2>Student Record</h2>

  <p>Roll No : @Model.RollNo</p>
  <p>Name : @Model.Name</p>
</div>

```

Output:

HTML_in_offline

- [Home](#)
- [Privacy](#)

Welcome

Learn about [building Web apps with ASP.NET Core](#).

Student Record

Roll No : 101

Name : Rahul

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HTML_in_offline

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Welcome

Learn about [building Web apps with ASP.NET Core](#).

Offline Page

Student Record

Roll No : 101

Name : Rahul

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